Chapter 2. Outliers and Robust Prediction

Predictive Modeling, Section 2 (MCD, LOF, Isolation Forest, leverage, Cook's distance, Huber, RANSAC, Theil–Sen)

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1 An Outlier Is Not the Same as a Data Error

What the observation is	Example	First response
input / measurement error	wrong unit, mis-recorded target	correct from the raw source; remove only if uncorrectable
a genuine rare case	a very large transaction, a rare high-risk patient	keep it, and check that group's prediction error separately
a case from a new environment	a policy or market condition absent in training	check distribution shift and the need to retrain

An outlier is relative to a reference; the action depends on why it is unusual.

1.1 Three kinds of outliers

Kind	Meaning	Simple diagnostic
univariate	one variable is far from most observations	histogram, boxplot, IQR rule
multivariate	each variable is ordinary but the *combination* is rare	scatterplot, multivariate distance, density
conditional	given x_i, the target y_i differs greatly from what is expected	residual $e_i = y_i - \hat{y}_i$

IQR rule: flag $x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$, $\mathrm{IQR} = Q_3 - Q_1$; it misses outliers in combinations.

Method	What it checks	Observations it looks for
MCD	global distance in X	multivariate outliers far from the central distribution
LOF	local density in X	points less dense than their neighbors
Isolation Forest	isolation path in X	points isolated by few random splits
studentised residual	y given X	cases far from the regression prediction
leverage, Cook's distance	influence on the regression fit	cases that change coefficients and predictions a lot

"X is unusual" and "y is unusual given X" are different problems.

2 MCD: Robust Mahalanobis Distance

2.1 Classical Mahalanobis distance and masking

With the sample mean $\bar{x}$ and sample covariance S,

$$D_i^2 = (x_i - \bar{x})^\top S^{-1} (x_i - \bar{x}).$$

Theorem: null distribution of the squared distance

If $X \sim N_p(\mu, \Sigma)$ then $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_p$. Hence, approximately, a point with $D_i^2 > \chi^2_{p,1-\alpha}$ is an outlier candidate.

Proof [Supplement]. $Z = \Sigma^{-1/2}(X-\mu) \sim N_p(0, I_p)$ and $(X-\mu)^\top \Sigma^{-1}(X-\mu) = Z^\top Z = \sum_{j=1}^p Z_j^2$. ■

Masking. Outliers pull $\bar{x}$ and inflate S, shrinking their own distances; estimate location and scatter from the bulk only.

2.2 Minimum Covariance Determinant

Definition: MCD

For a fixed h with $n/2 \le h < n$ (typically $h \approx 0.75n$), find the subset H of h observations whose sample covariance has the smallest determinant:

$$H^\star = \arg\min_{|H|=h} \det S_H, \qquad \hat{\mu}_{\text{MCD}} = \bar{x}_{H^\star}, \quad \widehat{\Sigma}_{\text{MCD}} = c_{h,n}\, S_{H^\star},$$

($c_{h,n}$ is a consistency factor), and use the robust distance

$$D_i^2 = (x_i - \hat{\mu}_{\text{MCD}})^\top \widehat{\Sigma}_{\text{MCD}}^{-1} (x_i - \hat{\mu}_{\text{MCD}}), \qquad \text{flag if } D_i^2 > \chi^2_{p,1-\alpha}.$$

$\det S_H \propto$ squared ellipsoid volume, so MCD finds the *tightest* h points; breakdown point $\approx (n-h)/n$. FAST-MCD iterates C-steps (replace H by the h points of smallest distance), which never increase $\det S_H$ (Exercise 2.2).

Remark: when MCD fits

Continuous data, $p < n$, one elliptical bulk; otherwise use LOF or Isolation Forest.

3 LOF: Lower Density Than the Neighbors

Definition: local outlier factor

Let $N_k(i)$ be the k nearest neighbors of i, $d_k(j)$ the distance from j to its kth neighbor, and the reachability distance $\text{rd}_k(i,j) = \max\{d_k(j), d(i,j)\}$. The local reachability density and LOF are

$$\text{lrd}_k(i) = \Big(\frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \text{rd}_k(i,j)\Big)^{-1}, \qquad \text{LOF}_k(i) = \frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \frac{\text{lrd}_k(j)}{\text{lrd}_k(i)}.$$

LOF ≈ 1: density like the neighbors; LOF $\gg 1$: local outlier. The comparison is *relative to the neighborhood*, so sparse genuine clusters are not penalized; rd_k stabilizes very close pairs.

4 Isolation Forest: Quickly Isolated Points

Algorithm: Isolation Forest

1. On a subsample, choose a feature uniformly at random.
2. Choose a split point uniformly between the feature's minimum and maximum.
3. Split recursively until each observation is alone (or a depth limit).
4. Grow many such trees and average the path lengths.

Anomaly score:

$$s(x) = 2^{-\mathbb{E}[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad H(m) \approx \ln m + 0.5772,$$

where $h(x)$ is the path length isolating x and $c(n)$ the average unsuccessful-search path length in a binary search tree of n points.

$s \to 1$: anomalous; $s \approx 0.5$ everywhere: no clear anomaly; $s \ll 0.5$: normal. `contamination` only sets the share labeled anomalous; the score cannot tell an error from a valid rare case.

5 Regression: Residual, Leverage and Influence

Motivation. Residuals flag bad fits, leverage flags unusual inputs, and influence combines both.

5.1 Three different notions

Suppose most data follow $y \approx 2x$.

Observation	Feature x	Residual	Interpretation
$(3, 20)$	ordinary range	large	outlier in the target
$(10, 20)$	far from the rest	small	high-leverage point that follows the trend
$(10, 0)$	far from the rest	large	influential point likely to change the line

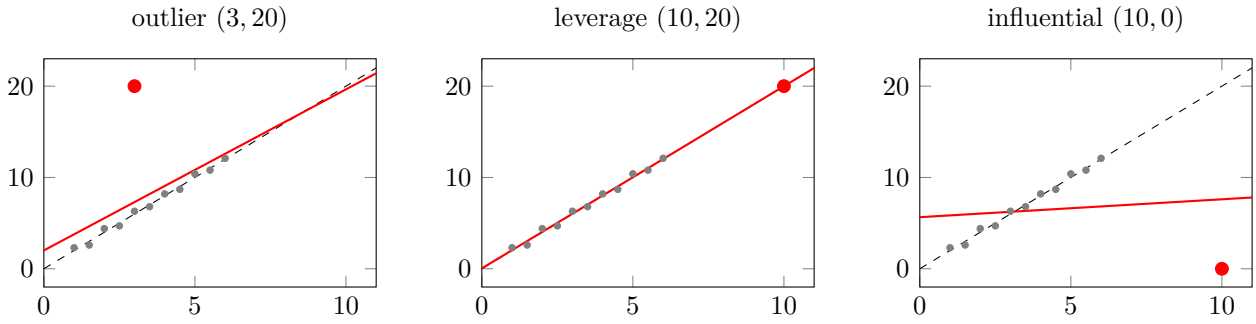


Figure 1: Dashed: OLS on the gray points; red: refit with the red point. Only the high-leverage point off the trend drags the slope (large Cook's distance).

5.2 Hat matrix and leverage

With $X \in \mathbb{R}^{n \times (p+1)}$ (intercept included) and $H = X(X^\top X)^{-1}X^\top$, $\hat{y} = Hy$ and $e = (I - H)y$. The leverage is $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$.

Theorem: properties of leverage and residuals

1. $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = p+1$, so the average leverage is $(p+1)/n$.
2. $\hat{y}_i = \sum_j h_{ij} y_j$, so $h_{ii} = \partial \hat{y}_i / \partial y_i$.
3. Under $\mathrm{Var}(\varepsilon) = \sigma^2 I$, $\mathrm{Var}(e_i) = \sigma^2(1-h_{ii})$.

Proof [Supplement]. $H = H^\top = H^2$: $h_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2 \Rightarrow 0 \le h_{ii} \le 1$; $\sum_i h_{ii} = \mathrm{tr}\{(X^\top X)^{-1} X^\top X\} = p+1$. Item 2 is $\hat{y} = Hy$. Item 3: $(I-H)X = 0 \Rightarrow e = (I-H)\varepsilon$, $\mathrm{Var}(e) = \sigma^2(I-H)$. ■

High-leverage points have small *raw* residuals (item 3), so standardize.

5.3 Studentised residual and Cook's distance

Diagnostic	Computation	Cases it reveals
studentised residual	$r_i = \dfrac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$	y_i far from what x_i predicts
leverage	h_{ii}, diagonal of H	far from the other observations in feature space
Cook's distance	$D_i = \dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$	removing it changes coefficients and predictions much

Screening rules $|r_i| > 2$, $h_{ii} > 2(p+1)/n$, $D_i > 4/n$ trigger *inspection*, not deletion.

Theorem: leave-one-out update and Cook's formula

Let $\hat{\beta}_{(i)}$ be the OLS estimate without case i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^\top X)^{-1} x_i e_i}{1-h_{ii}},$$

and Cook's distance, defined as $D_i = \dfrac{\|X\hat{\beta} - X\hat{\beta}_{(i)}\|^2}{(p+1)\hat{\sigma}^2}$, equals $\dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$.

Proof [Supplement]. $A = X^\top X$, $A_{(i)} = A - x_i x_i^\top$, $X_{(i)}^\top y_{(i)} = X^\top y - x_i y_i$; Sherman–Morrison: $A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}$. Then

$$\hat{\beta}_{(i)} = \left(A^{-1} + \tfrac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}\right)(X^\top y - x_i y_i) = \hat{\beta} - A^{-1}x_i y_i + \frac{A^{-1}x_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i(y_i - \hat{y}_i)}{1-h_{ii}}.$$

So $\|X(\hat{\beta} - \hat{\beta}_{(i)})\|^2 = \frac{e_i^2}{(1-h_{ii})^2} x_i^\top A^{-1} A A^{-1} x_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}$; divide by $(p+1)\hat{\sigma}^2$ and use $e_i^2 = r_i^2\hat{\sigma}^2(1-h_{ii})$. ■

D_i is large only if *both* r_i and $h_{ii}/(1-h_{ii})$ are, as for $(10, 0)$.

Remark: leave-one-out residuals for free

$y_i - x_i^\top \hat{\beta}_{(i)} = e_i/(1-h_{ii})$, so OLS LOOCV error is $\frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ without refitting.

6 What to Do With Detected Cases

Finding	Action	Caveat
input / measurement error	correct or remove in raw data	remove only if it cannot be corrected and cannot be used
heavily skewed variable	log or Box–Cox transform	keeps the order, compresses gaps among large values
extremes must be capped	winsorisation	cap both tails at quantiles *estimated on training data*
valid rare case	keep	use robust learning and segment-wise evaluation

Never delete or alter observations on a detection score alone.

7 Robust Learning

Method	Core principle	Suitable when
Huber regression	caps the growth rate of loss for large residuals	a few large target outliers
RANSAC	fits many small subsamples; keeps the model supported by most points	clear inlier vs. contaminated groups
Theil–Sen	median of pairwise slopes	linear regression with few features

7.1 Huber loss

With residual $r = y - \hat{y}$:

Loss	Definition	Large residual	Target of prediction
squared	r^2	grows quadratically	conditional mean
absolute	$\lvert r\rvert$	grows linearly	conditional median
Huber	quadratic for small, linear for large $\lvert r\rvert$	linear beyond δ	compromise between center fit and tail robustness

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\left(|r| - \frac{1}{2}\delta\right), & |r| > \delta, \end{cases} \qquad \psi_\delta(r) = \rho'_\delta(r) = \begin{cases} r, & |r| \le \delta, \\ \delta\,\mathrm{sign}(r), & |r| > \delta. \end{cases}$$

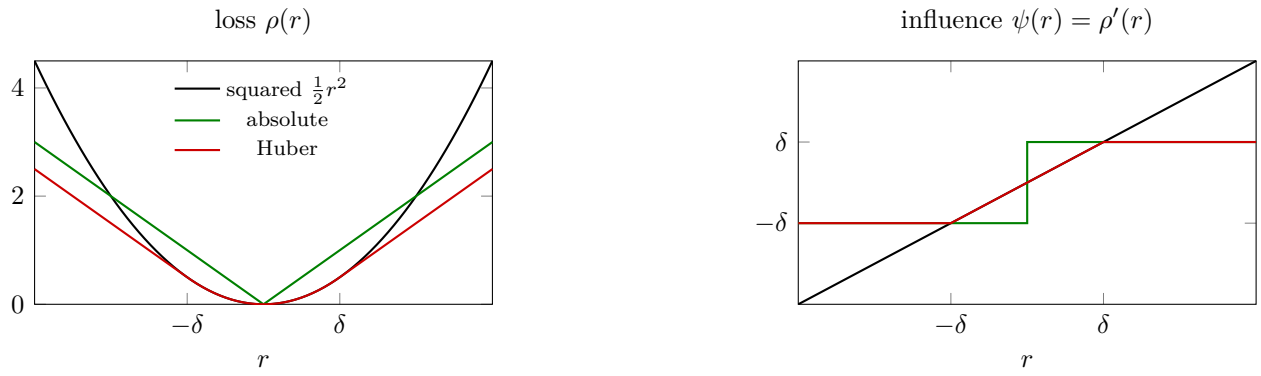


Figure 2: Losses (left) and influence functions (right): unbounded for squared loss, capped at $\pm\delta$ for Huber.

Derivation: why Huber limits outliers — IRLS view

Estimating equation $\sum_i \psi_\delta(y_i - x_i^\top \beta)x_i = 0$; with $\psi_\delta(r) = w(r)\,r$, $w(r) = \min\{1, \delta/|r|\}$, iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$, $W = \mathrm{diag}\{w(r_i)\}$. A residual 10δ gets weight 0.1; influence is bounded by δ. ρ_δ is convex and C^1; $\delta \to \infty$ gives OLS, $\delta \to 0$ (rescaled) LAD.

Usually $\delta = 1.345\hat{s}$, $\hat{s} = \mathrm{MAD}/0.6745$ (95% Gaussian efficiency).

7.2 RANSAC

Algorithm: RANSAC

1. Fit the regression on a small random subsample (minimal size s).
2. Classify observations with residual below a threshold as inliers.
3. Repeat many times.
4. Refit the model on the largest inlier set.

With inlier fraction w, success at least once with probability q needs $N \geq \log(1-q)/\log(1-w^s)$ trials.

7.3 Theil–Sen

In simple regression compute all pairwise slopes:

$$\hat{\beta}_1 = \underset{i<j}{\mathrm{median}} \frac{y_j - y_i}{x_j - x_i}, \qquad \hat{\beta}_0 = \underset{i}{\mathrm{median}}(y_i - \hat{\beta}_1 x_i).$$

Breakdown point $\approx 1 - 1/\sqrt{2} \approx 29\%$.

8 Robust Prediction Is Also Chosen by CV

Algorithm: protocol

1. Split off the final test set first.
2. Inside each CV training fold, fit scaling and outlier-detection steps.
3. Set the detection threshold, LOF's k, Huber's δ, RANSAC's residual threshold from training folds only.
4. Compare OLS and robust regressions on the same validation folds.
5. Check errors of outlier candidates and of rare groups along with the overall error.

Result to check	Purpose
MAE and RMSE	evaluate typical errors and rare large errors together
sensitivity to the detection rule	does model selection change with the threshold?
error of rare groups	did robustification ignore important cases?

Summary

- Outliers are relative to a reference; errors, valid rare cases and new environments call for different actions.
- In X: MCD (global, elliptical), LOF (local density), Isolation Forest (isolation depth).
- In regression: residual (y given x), leverage (x unusual), Cook's distance (both together).
- Robust losses (Huber, LAD) and robust fitting (RANSAC, Theil–Sen) limit the dominance of a few points; their tuning constants are chosen inside CV.

9 Selected exercises

Motivation. Three facts used above: affine invariance, C-step monotonicity, and Huber as shifted least squares.

Exercise 2.1 (Affine invariance of Mahalanobis distances)

Let $x_1, \dots, x_n \in \mathbb{R}^p$ have sample covariance $S \succ 0$, and let $y_i = Ax_i + b$ with A an invertible $p \times p$ matrix and $b \in \mathbb{R}^p$.
(a) Show that the squared Mahalanobis distances D_i^2 computed from the y_i equal those computed from the x_i.
(b) Show that the MCD subset $H^\star$ is the same for both data sets, and so are the robust distances.

Solution [Supplement].

(a)

Step 1. $\bar{y} = A\bar{x} + b$ and $y_i - \bar{y} = A(x_i - \bar{x})$, so $S_y = AS_xA^\top$, which is invertible with $S_y^{-1} = A^{-\top}S_x^{-1}A^{-1}$.

Step 2. Hence

$$(y_i - \bar{y})^\top S_y^{-1}(y_i - \bar{y}) = (x_i - \bar{x})^\top A^\top A^{-\top} S_x^{-1} A^{-1} A(x_i - \bar{x}) = (x_i - \bar{x})^\top S_x^{-1}(x_i - \bar{x}).$$

(b)

Step 3. For any H, $S_H^{(y)} = AS_H^{(x)}A^\top$, so $\det S_H^{(y)} = (\det A)^2 \det S_H^{(x)}$ with a common factor $(\det A)^2 > 0$: same minimizing subsets.

Step 4. Then $\hat{\mu}^{(y)} = A\hat{\mu}^{(x)} + b$, $\widehat{\Sigma}^{(y)} = A\widehat{\Sigma}^{(x)}A^\top$ ($c_{h,n}$ common), and Step 2 gives equal robust distances.

LOF and Isolation Forest lack this invariance, so scale their inputs deliberately. ■

Exercise 2.2 (A C-step never increases the determinant)

For a subset H with $|H| = h$ put $T_H = \frac{1}{h}\sum_{i \in H} x_i$ and $S_H = \frac{1}{h}\sum_{i \in H}(x_i - T_H)(x_i - T_H)^\top$. Let $|H_1| = h$ with $S_1 = S_{H_1} \succ 0$, $T_1 = T_{H_1}$, and $d_1(i) = (x_i - T_1)^\top S_1^{-1}(x_i - T_1)$. Let H_2 consist of the h indices with the smallest $d_1(i)$, and $S_2 = S_{H_2}$. Show that $\det S_2 \le \det S_1$.

Solution [Supplement].

Step 1. Using $a^\top M a = \operatorname{tr}(Maa^\top)$,

$$\sum_{i\in H_1} d_1(i) = \operatorname{tr}\Big(S_1^{-1}\sum_{i\in H_1}(x_i - T_1)(x_i - T_1)^\top\Big) = \operatorname{tr}(S_1^{-1} h S_1) = hp.$$

Step 2. $\sum_{i\in H_2} d_1(i) \le \sum_{i\in H_1} d_1(i) = hp$ by the choice of H_2.

Step 3. For $M \succ 0$ and any c, since $\sum_{i\in H_2}(x_i - T_2) = 0$ with $T_2 = T_{H_2}$,

$$\sum_{i\in H_2}(x_i - c)^\top M(x_i - c) = \sum_{i\in H_2}(x_i - T_2)^\top M(x_i - T_2) + h(T_2 - c)^\top M(T_2 - c).$$

With $M = S_1^{-1}$, $c = T_1$: left side $= \sum_{i\in H_2} d_1(i)$, first right term $= h\operatorname{tr}(S_1^{-1}S_2)$, so $\operatorname{tr}(S_1^{-1}S_2) \le p$ by Step 2.

Step 4. $S_1^{-1/2} S_2 S_1^{-1/2} \succeq 0$ has eigenvalues $\lambda_j \ge 0$, trace $\le p$, determinant $\det S_2/\det S_1$. By AM–GM,

$$\Big(\frac{\det S_2}{\det S_1}\Big)^{1/p} = \Big(\prod_j \lambda_j\Big)^{1/p} \le \frac{1}{p}\sum_j \lambda_j \le 1.$$

With finitely many h-subsets the non-increasing sequence stabilizes, where FAST-MCD stops. (Normalizing by $h-1$ rescales all determinants equally.) ■

Exercise 2.3 (Huber regression is least squares with outlier shifts)

(a) For $r \in \mathbb{R}$ and $\delta > 0$ show that

$$\rho_\delta(r) = \min_{\gamma\in\mathbb{R}} \Big\{\tfrac{1}{2}(r-\gamma)^2 + \delta|\gamma|\Big\},$$

with the unique minimizer $\gamma^\star(r) = \operatorname{sign}(r)(|r| - \delta)_+$.
(b) Let $F(\boldsymbol{\beta}, \gamma) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta} - \gamma\|_2^2 + \delta\|\gamma\|_1$ with $\gamma \in \mathbb{R}^n$. Show that $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F if and only if $\hat{\boldsymbol{\beta}}$ minimizes $\sum_i \rho_\delta(y_i - x_i^\top\boldsymbol{\beta})$ and $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top\hat{\boldsymbol{\beta}})$ for all i.

Solution [Supplement].

(a)

Step 1. $\phi(\gamma) = \frac{1}{2}(r-\gamma)^2 + \delta|\gamma|$ is strictly convex and coercive; its unique minimizer solves $0 \in \gamma - r + \delta\,\partial|\gamma|$ ($\partial|\gamma| = \{\operatorname{sign}\gamma\}$ for $\gamma \ne 0$, $[-1,1]$ at 0).

Step 2. $|r| \le \delta$: $\gamma = 0$ ($r \in \delta[-1,1]$), $\phi(0) = \frac{1}{2}r^2$. $|r| > \delta$: $\gamma = r - \delta\operatorname{sign}(r)$ satisfies $\gamma - r + \delta\operatorname{sign}\gamma = 0$, value $\frac{1}{2}\delta^2 + \delta(|r| - \delta) = \delta(|r| - \frac{1}{2}\delta)$. Both give $\rho_\delta(r)$ at $\gamma^\star(r)$.

(b)

Step 3. With $r_i = y_i - x_i^\top\boldsymbol{\beta}$, $F = \sum_i\{\frac{1}{2}(r_i - \gamma_i)^2 + \delta|\gamma_i|\}$ separates, so by (a) $\min_\gamma F(\boldsymbol{\beta}, \gamma) = \sum_i \rho_\delta(r_i) =: \mathcal{H}(\boldsymbol{\beta})$, attained only at $\gamma_i = \gamma^\star(r_i)$.

Step 4. If $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F, then for every $\boldsymbol{\beta}$

$$\mathcal{H}(\hat{\boldsymbol{\beta}}) \le F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) \le \min_\gamma F(\boldsymbol{\beta}, \gamma) = \mathcal{H}(\boldsymbol{\beta}),$$

so $\hat{\boldsymbol{\beta}}$ minimizes $\mathcal{H}$; $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}})$, and uniqueness forces $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top\hat{\boldsymbol{\beta}})$. Conversely, $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}}) \le \mathcal{H}(\boldsymbol{\beta}) \le F(\boldsymbol{\beta}, \gamma)$ for all $(\boldsymbol{\beta}, \gamma)$.

Step 5. Given $\hat{\gamma}$, F is quadratic in $\boldsymbol{\beta}$: $X^\top(\mathbf{y} - \hat{\gamma} - X\hat{\boldsymbol{\beta}}) = 0$, OLS on $\mathbf{y} - \hat{\gamma}$.

Cases with $|r_i| \le \delta$ get no shift; others are shifted by $|r_i| - \delta$, so only $\pm\delta$ reaches the fit (bounded influence). ■